Analyzing the impact of increasing the value of k in the A* algorithm

The Estimate shown in the tables represents the value of each coefficient. Positive estimates indicate an increase in the response variable, while negative estimates indicate a decrease. Furthermore, the greater the magnitude of the estimate (i.e., the absolute value), the stronger the effect on the response variable, whether it is an increase or a decrease. The standard error (Std. Error) associated with each estimate indicates the precision of this estimate; the smaller the standard error, the greater the accuracy. The t value is calculated by dividing the estimate by the standard error and indicates statistical significance: high values (in absolute terms) suggest that the estimate is robust. The p-value complements this analysis, indicating whether the estimate is statistically significant at the defined significance level of $\alpha = 0.01$.

For example, the average computing time estimate for the A*-k4 technique (Table 1) is -0.348577. Applying the exponential function (since the logarithm was used as the link function), it follows that $\exp(-0.348577) = 0.71$. This indicates that the A*-k4 requires 0.71 times the computing time of the A*-k3 baseline method for the same path distances. In contrast, the A*-k5 method requires 2.69 times more computing time than the A*-k3 method.

All the p-values obtained from the regression models for the initial experiments are below the significance level of $\alpha = 0.01$. Therefore, all the results are statistically significant.

Table 1: Statistical data of the impact of k on Complex map: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	0.026110	0.001483	17.603	<2e-16
A*-k4	-0.348577	0.085390	-4.082	4.59e-05
A*-k5	0.989686	0.090268	10.964	<2e-16

Table 2: Statistical data of the impact of k on Complex map: Number of nodes in the resulting path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	0.0164483	0.0002064	79.67	<2e-16
A*-k4	-0.3957170	0.0118847	-33.30	<2e-16
A*-k5	-0.7585925	0.0125636	-60.38	<2e-16

Table 3: Statistical data of the impact of k on StdA maps: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	0.0095252	0.0002199	43.315	<2e-16
A*-k4	-0.9222809	0.0763222	-12.084	<2e-16
A*-k5	0.3658363	0.0786007	4.654	3.28e-06

Table 4: Statistical data of the impact of k on StdA maps: Number of nodes in the resulting path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	4.799e-03	2.297e-05	208.90	<2e-16
A*-k4	-3.975e-01	7.973e-03	-49.86	<2e-16
A*-k5	-6.956e-01	8.211e-03	-84.72	<2e-16

Table 5: Statistical data of the impact of k on StdB maps: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	0.0072077	0.0001766	40.806	<2e-16
A*-k4	0.6107196	0.0678985	8.995	<2e-16
A*-k5	2.4326089	0.0709140	34.304	<2e-16

Table 6: Statistical data of the impact of k on StdB maps: Number of nodes in the resulting path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	3.688e-03	2.176e-05	169.52	<2e-16
A*-k4	-4.283e-01	8.364e-03	-51.21	<2e-16
A*-k5	-7.127e-01	8.735e-03	-81.58	<2e-16

Final results

All the p-values obtained from the regression models for the final experiments are below the significance level of $\alpha = 0.01$. Therefore, it can be concluded that all results are statistically significant.

Table 7: Statistical data of the Complex map: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-k3	0.022190	0.001024	21.678	<2e-16
A*-k4	-0.376494	0.087687	-4.294	1.79e-05
T*-k3	0.268862	0.086425	3.111	0.00187
T*-k4	-0.415880	0.086512	-4.807	1.57e-06
JPS-k3	0.824114	0.086370	9.542	<2e-16
JPS-k4	3.992990	0.086383	46.224	<2e-16

Table 8: Statistical data of the Complex map: Number of opened nodes.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	0.0227662	0.0008395	27.120	<2e-16
A*-$k4$	-0.7746296	0.0719123	-10.772	<2e-16
T*-$k3$	-0.8275522	0.0708769	-11.676	<2e-16
T*-$k4$	-3.0816508	0.0709485	-43.435	<2e-16
JPS-$k3$	-0.7447971	0.0708320	-10.515	<2e-16
JPS-$k4$	-0.3756777	0.0708425	-5.303	1.18e-07

Table 9: Statistical data of the Complex map: Number of nodes in the result path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	0.0147722	0.0001556	94.95	<2e-16
A*-$k4$	-0.3994847	0.0133269	-29.98	<2e-16
T*-$k3$	-0.7011036	0.0131350	-53.38	<2e-16
T*-$k4$	-0.7570673	0.0131483	-57.58	<2e-16
JPS-$k3$	-0.5661504	0.0131267	-43.13	<2e-16
JPS-$k4$	-0.6198915	0.0131286	-47.22	<2e-16

Table 10: Statistical data of the StdA maps: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	0.0075102	0.0001714	43.808	<2e-16
A*-$k4$	-1.0947727	0.0871829	-12.557	<2e-16
T*-$k3$	-0.2394544	0.0863593	-2.773	0.00556
T*-$k4$	-0.4244838	0.0863611	-4.915	8.92e-07
JPS-$k3$	2.2510557	0.0865033	26.023	<2e-16
JPS-$k4$	3.7867018	0.0871884	43.431	<2e-16

Table 11: Statistical data of the StdA maps: Number of opened nodes.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	0.0080677	0.0000963	83.78	<2e-16
A*-$k4$	-2.0008415	0.0489745	-40.85	<2e-16
T*-$k3$	-2.2753890	0.0485119	-46.90	<2e-16
T*-$k4$	-4.1662679	0.0485129	-85.88	<2e-16
JPS-$k3$	-1.5701851	0.0485927	-32.31	<2e-16
JPS-$k4$	-1.8315883	0.0489776	-37.40	<2e-16

Table 12: Statistical data of the StdA maps: Number of nodes in the resulting path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	4.415e-03	1.831e-05	241.11	<2e-16
A*-$k4$	-3.998e-01	9.313e-03	-42.93	<2e-16
T*-$k3$	-6.203e-01	9.225e-03	-67.24	<2e-16
T*-$k4$	-8.058e-01	9.225e-03	-87.35	<2e-16
JPS-$k3$	-8.125e-01	9.240e-03	-87.93	<2e-16
JPS-$k4$	-7.744e-01	9.313e-03	-83.15	<2e-16

Table 13: Statistical data of the StdB maps: Execution time.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	0.0068543	0.0001359	50.454	<2e-16
A*-$k4$	0.5665870	0.0733897	7.720	1.21e-14
T*-$k3$	0.4985187	0.0733618	6.7959	1.11e-11
T*-$k4$	-1.0910391	0.0731142	-14.922	<2e-16
JPS-$k3$	1.0281518	0.0735096	13.987	<2e-16
JPS-$k4$	3.5444352	0.0854672	41.471	<2e-16

Table 14: Statistical data of the StdB maps: Number of opened nodes.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	6.487e-03	9.056e-05	71.64	<2e-16
A*-$k4$	-2.341e+00	4.892e-02	-47.86	<2e-16
T*-$k3$	-2.822e+00	4.890e-02	-57.70	<2e-16
T*-$k4$	-5.170e+00	4.874e-02	-106.08	<2e-16
JPS-$k3$	-2.750e+00	4.900e-02	-56.13	<2e-16
JPS-$k4$	-2.697e+00	5.697e-02	-47.34	<2e-16

Table 15: Statistical data of the StdB maps: Number of nodes in the resulting path.

Algorithm	Estimate	Std. Error	t value	P(> |t |)
A*-$k3$	3.805e-03	1.809e-05	210.41	<2e-16
A*-$k4$	-4.265e-01	9.770e-03	-43.65	<2e-16
T*-$k3$	-6.160e-01	9.766e-03	-63.07	<2e-16
T*-$k4$	-8.127e-01	9.733e-03	-83.50	<2e-16
JPS-$k3$	-8.473e-01	9.786e-03	-86.59	<2e-16
JPS-$k4$	-7.722e-01	1.138e-02	-67.87	<2e-16